

DS 202: Introduction to Data Structures and Algorithms

Course Code	DS 202
Course Name	Introduction to Data Structures and Algorithms
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	UG
Distribution	Core for B.Tech Data Science and Engineering, Not allowed to B.Tech. CSE, Discipline elective for B.Tech in EE, Free elective for other B.Tech disciplines.
Semester	Even/Odd
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019), in the B.Tech Data Science and Engineering curriculum

Preamble

Data science uses the concepts from statistics and machine learning principles to write code such that one can effectively explore the problem at hand. This necessitates at least a basic understanding of data structures, algorithms, and time-space complexity so that one can program more efficiently and understand the tools used. With larger datasets, this becomes particularly important. The way that code is written influences the speed at which data is analysed and conclusions can be reached accordingly. The objective of the course is to expose students to relevant data structures and algorithms that relate to time complexity. At the end students will learn how to write efficient codes using suitable data structures and can design algorithms for practical data science related problems.

Course Modules

- **Complexity Analysis:** Time and Space complexity of algorithms, asymptotic analysis, average and worst case analysis, asymptotic notation, importance of efficient algorithms, program performance measurement, data structures and algorithms. (2 hours)
- **Stacks, Queues and Lists:** Abstract data types, sequential and linked implementations of stacks, queues and lists, comparison of insertion, deletion and search operations for sequential and linked lists, list and chain classes, doubly linked lists, circular lists, applications of lists in bin sort, radix sort, sparse tables. (6 hours)
- **Hashing:** Hashing as a search structure, hash table, collision resolution, universal hashing, linear open addressing, chains, hash tables in data-compression, LZW algorithm. (3 hours)
- **Trees and Search Trees:** Abstract data type, sequential and linked implementations, tree traversal methods and algorithms, Binary trees and their properties, Binary search trees, search efficiency, insertion and deletion operations, importance of balancing, AVL trees, searching, insertion and deletions in AVL trees, Tries, 2-3 tree, B-tree. (6 hours)
- **Heaps:** Heaps as priority queues, heap implementation, insertion and deletion operations, binary heaps, binomial and Fibonacci heaps, heapsort, heaps in Huffman coding. (3 hours)
- **Graphs:** Definition, terminology, directed and undirected graphs, properties, implementation – adjacency matrix and linked adjacency chains, connectivity in graphs, graph traversal – breadth first and depth first, spanning trees. (3 hours)
- **Basic Algorithmic Techniques:** Greedy algorithms, divide & conquer, dynamic programming. Search techniques – backtracking, Sorting algorithms with analysis, integer sorting, selection sort. Graph algorithms: DFS and BFS with applications, MST and shortest paths. (5 hours)

Laboratory Modules

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be appropriately be sequenced for synchronization with the laboratory.

Text Books

1. Cormen, T. H., Leiserson, C. E., Rivest, R. L., and Stein C., *Introduction to Algorithms*, MIT Press, 2009.

References

1. Sahni, S., *Data Structures, Algorithms, and Applications in C++*, Silicon Press, 2005.
2. Tenenbaum, A. M., Langsam Y., and Augenstein, M. J., *Data Structures Using C and C++*, Prentice Hall, 1995.